STOCHASTIC METHODS IN WATER RESOURCES

Unit 3: Random functions
Lecture 10: Spectral representation of random functions

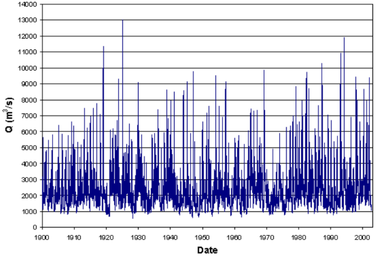
Luis Alejandro Morales, Ph.D.

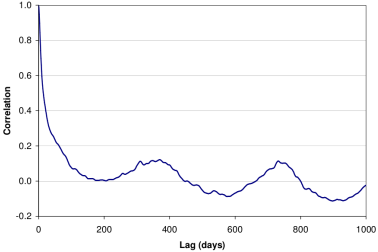
Universidad Nacional de Colombia
Department of Civil and Agriculture Engineering

October 14, 2025

Spectral representation of random functions: Definitions

- Correlation functions of a time series, when analyzed for a longer period exhibit a periodic behaviour e.g. Hole effect model (see tables in lecture 9).
- For instance, the figures below show a time series of daily average streamflows and its correlation function.





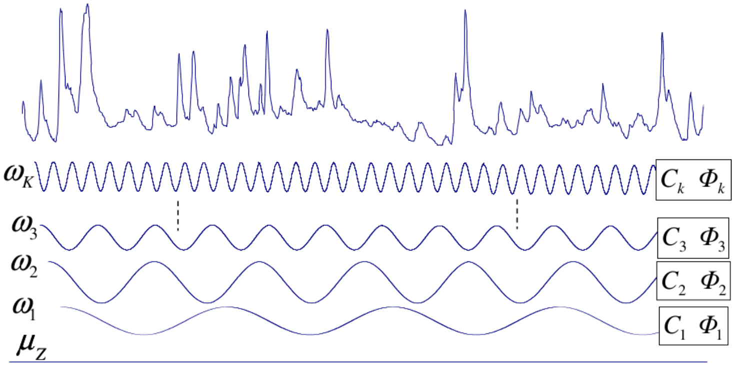
- A clear periodic behaviour is observed in the correlation function which is also apparent in the time series.
- Such periodicity is caused because evaporation, which is driven by radiation and temperature, has a clear seasonal character, specially, at higher and lower latitudes and temperate climates.
- In conclusion, most hydrological time series show a seasonal variation. This means that to analyse these models with stationary random functions requires that seasonality must be removed.
- The occurrence of seasonality has also triggered the use of **spectral methods** in stochastic modelling, although it must be stressed that spectral methods are also very suitable for analysing stationary random functions.

Spectral density function

The spectral representation of a stationary random function $Z(t)$, means that $Z(t)$ is composed by the sum of its mean μ_Z and $2K$ **sinosoids** with increasing frequencies, where each frequency has a **random amplitude** C_K and a **random phase angle** Φ_K. So $Z(t)$ is expressed as

$$Z(t) = \mu_Z + \sum_{k=-K}^{K} C_K \cos(\omega_K t + \Phi_K)$$

where $C_K = C_{-K}$, $\Phi_K = \Phi_{-K}$ and $\omega_K = \pm\frac{1}{2}(2k-1)\Delta\omega$.
The figure below is a schematic representation of the spectral decomposition of a stationary random function. The random signal is decomposed into its harmonics of increasing frequency (ω_K) with random amplitude C_K and random phase angle Φ_K.



$$Z(t) = \mu_Z \pm C_1\cos(\omega t + \Phi_1) \pm C_2\cos(2\omega t + \Phi_2) \pm \cdots \pm C_K\cos(K\omega t + \Phi_K)$$

Spectral density function

Based on the mathematical representation of the spectral decomposition of the random process, the following relationships known as the **Winer-Khinchine relations** are obtained

$$C_Z(\tau) = \int_{-\infty}^{\infty} S_Z(\omega)\cos(\omega\tau)d\omega$$
$$S_Z(\omega) = \frac{1}{2\pi}\int_{-\infty}^{\infty} C_Z(\tau)\cos(\omega\tau)d\tau$$

where the function $S_Z(\omega)$ is known as the **spectral density function** of the random process. Note that the equations above show that covariance function C_Z is a **Fourier transform** of the spectrum and viceversa. These two equations form a Fourier pair. Making the lag $\tau = 0$, it is possible to understand the physical meaning of $S_Z(\omega)$ because

$$C_Z(0) = \int_{-\infty}^{\infty} S_Z(\omega)d\omega$$

from where it can be seen that the variance of the random function is equal to the integral over the spectral density.This means that the variance is a weighted sum of variance components, where each component consists of a random harmonic function of a given frequency.The spectral density then represents the weight of each of the attributing random harmonics, i.e. the relative importance of each random harmonic in explaining the total variance of the random function.

Spectral density function

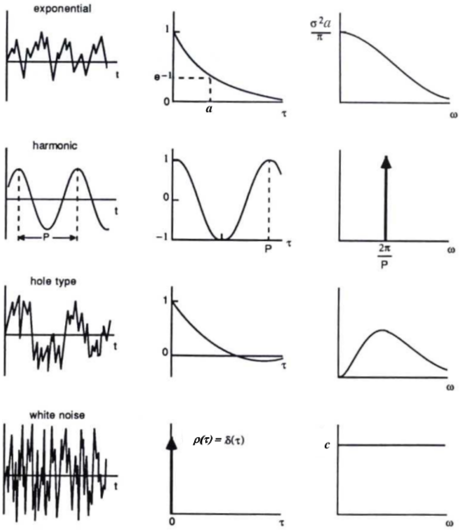
It is easy to see the analogy with the electromagnetic spectrum where the total energy of electromagnetic radiation (which is analogous to the variance of our random signal) is found by the area under the spectrum and can be attributed to relative contributions from different wavelengths.
The following table shows expressions for spectral density functions for different covariance functions.

Exponential covariance	$C_Z(\tau) = \sigma_Z^2 e^{-\lvert\tau\rvert/a}$ for $a > 0$	$S_Z(\omega) = \frac{a\sigma_Z^2}{\pi(1+a^2\omega^2)}$
Random harmonic: $Z(t) = a\cos(\omega_0 t + \phi)$ where a and ω_0 are deterministic constants and ϕ is random.	$C_Z(\tau) = \frac{a^2}{2}\cos(\omega_0\tau)$ for $\tau \geq 0$ and $a, \omega_0 > 0$	$S_Z(\omega) = \frac{a^2}{4}\delta(\omega - \omega_0)$
Hole effect (wave) model	$C_Z(\tau) = \sigma_Z^2(1 - \lvert t\rvert/a)e^{-\lvert\tau\rvert/a}$ for $a > 0$	$S_Z(\omega) = \frac{a^3\sigma_Z^2\omega^2}{\pi(1+a^2\omega^2)^2}$
White noise model	$\rho_Z(\tau) = \begin{cases} 1, \text{ if } \tau = 0 \\ 0 = , \text{ if } \tau > 0 \end{cases}$	$S_Z(\omega) = c$

Spectral density function

The following figure shows typical realisations of the random functions involved, their correlation function and the associated spectrum.

- Note that the spectrum of white noise is a horizontal line, implying an infinite variance according to the equation.
- This shows that white noise is a mathematical construct, and not a feasible physical process: the area under the spectrum is a measure for the total energy of a process.
- This area is infinitely large, such that all the energy in the universe would not be sufficient to generate such a process.
- In practice one often talks about wide band processes, where the spectrum has a wide band of frequencies, but encloses a finite area.



Spectral density function

Note that spectral density function is an even function as the covariance function: $S_Z(\omega) = S_Z(-\omega)$. This means that the spectral density function can be expressed as a **one-sided function** $G_Z(\omega) = 2S_Z(\omega)$, for $\omega \geq 0$. The Wiener-Khinchine relations become

$$C_Z(\tau) = \int_0^{\infty} G_Z(\omega)\cos(\omega\tau)d\omega$$
$$G_Z(\omega) = \frac{2}{\pi}\int_0^{\infty} C_Z(\tau)\cos(\omega\tau)d\omega$$

Sometimes it is convenient to work with the normalised spectral density functions, by dividing the spectra by the variance: $s_Z(\omega) = \frac{S_Z(\omega)}{\sigma_Z^2}$ and $g_Z(\omega) = \frac{G_Z(\omega)}{\sigma_Z^2}$. There is a relation between the normalised spectral density function $s_Z(\omega)$ and the scale of fluctuation. So if $\omega = 0$ we obtain

$$s_z(0) = \frac{1}{2\pi}\int_{-\infty}^{\infty} \rho_Z(\tau)d\omega = \frac{\theta}{2\pi}$$

Formal spectral representation

A more formal representation of the spectral density is used in the literature based on complex calculus. Here the random function $Z(t)$ is defined as the real part *Re.* of a complex random function $Z^*(t)$.

$$Z(t) = Re\{Z^*(t)\} = Re\left\{\mu_Z + \sum_{k=-K}^{K} X_k e^{i\omega_k t}\right\} \xrightarrow[K\to\infty]{} Re\left\{\mu_Z + \int_{-\infty}^{\infty} e^{i\omega t} dX(\omega)\right\}$$

where $\omega_k = \frac{1}{2}(2k-1)\Delta\omega$ is the frequency, and X_k a random complex number representing the amplitude. This equation means that the complex random process is decomposed into a large number of complex harmonic functions $e^{i\omega t} = \cos(\omega t) + i\sin(\omega t)$ with random complex amplitude. According to this representation, the Wiener-Khinchine equations are

$$C_Z(\tau) = \int_{-\infty}^{\infty} S_Z(\omega) e^{i\omega\tau} d\omega$$

$$S_Z(\omega) = \frac{1}{2\pi} \int_{-\infty}^{\infty} C_Z(\tau)^{i\omega\tau} d\tau$$

It can be demonstrated that these equations are equivalent to the prior Wiener-Khinchine the equations.

Estimating the spectral density function

For stationary random functions, the spectral density function can be estimated from the covariance function as

$$S_Z(\omega_i) = \frac{1}{2\pi}\left(\lambda_0 C_Z(0) + 2\sum_{k=1}^{M}\lambda_k \hat{C}_Z(k\Delta\tau)\cos(\omega_i k)\right)$$

where $\omega_i = i\pi/M$ and $|i| = 1, 2, \cdots, M$. The weights λ_k are necessary to smooth the covariances before performing the transformation. So that, a smooth spectral density function is obtained displaying only the relevant features. There are numerous ways to estimate the smoothing weights, the most common are:

- **Tukey window**

$$\lambda_k = \frac{1}{2}\left(1 + \cos\left(\frac{k\pi}{M}\right)\right), \qquad k = 1, 2, \cdots, M$$

- **Parzen window**

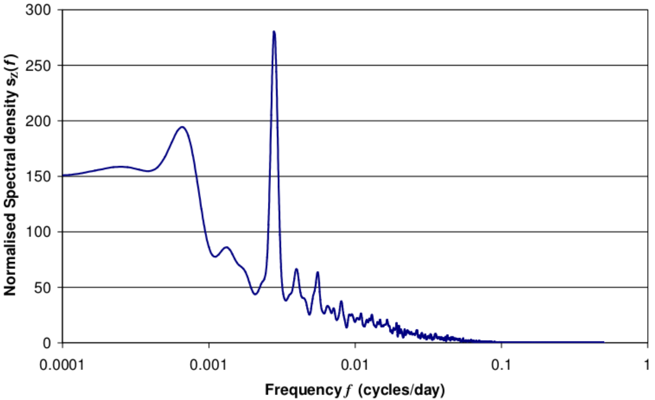
$$\lambda_k = \begin{cases} 1 - 6\left(\frac{k}{M}\right)^2 + 6\left(\frac{k}{M}\right)^3, & 0 \le k \le M/2 \\ \left(2 - \frac{k}{m}\right)^3, & M/2 \le k \le M \end{cases}$$

The highest frequency that is analysed is equal to $f_{max} = \frac{\omega_{max}}{2\pi} = 0.5$, which is the highest frequency that can be estimated from a time series: half of the frequency of the observations. This frequency is called the **Nyquist frequency**. For instance, if streamflow is observed once per day, the highest frequency that can be detected is on cycle per two days.

Estimating the spectral density function

The smallest frequency (largest wavelength) that can be analysed depends on the discretisation $f_{min} = \pi/(2\pi M) = 1/2M$, where M is the cutoff level (maximum lag considered) of the covariance function. The width of the smoothing windows (smoothing weight) is adjusted accordingly.

The figure below shows the spectrum of a daily discharge time series. The spectral density function was estimated using a Parzen window with $M = 9000$. Clearly, small frequencies dominate with a small peak between 4 and 5 years. Most prominent of course, as expected, there is a peak at a frequency of once a year, which exemplifies the strong seasonality in the time series.



Spectral representations of random space functions

Extending the equations above to two dimensions, the stationary random function $Z(x_1, x_2)$ can be expressed in terms of random harmonics as

$$Z(x_1, x_2) = \mu_Z + \sum_{k_1=-K_1}^{k_1=K_1} \sum_{k_2=-K_2}^{k_2=K_2} C_{12} \cos(\omega_{k_1} x_1 + \omega_{k_2} x_2 + \Phi_{12})$$

where C_{12} and Φ_{12} are the random amplitude and the phase angle, respectively, belonging to frequency ω_{k_i}, $i = 1, 2$

$$\omega_{k_i} = \pm \frac{2k_i \Delta\omega_i - 1}{2}$$

The Wiener-Khinchine equations then become

$$C_Z(h_1, h_2) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} S_Z(\omega_1, \omega_2) \cos(\omega_1 h_1 + \omega_2 h_2) d\omega_1 d\omega_2$$

$$S_Z(\omega_1, \omega_2) = \frac{1}{(2\pi)^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} C_Z(h_1, h_2) \cos(\omega_1 h_1 + \omega_2 h_2) d\omega_1 d\omega_2$$

The variance is given by the volume under the 2D spectral density function $S_Z(\omega_1, \omega_2)$. Therefore

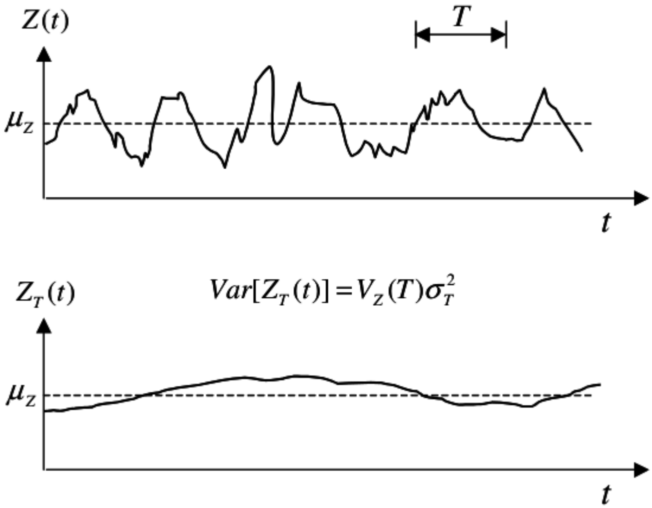
$$\sigma_Z^2 = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} S_Z(\omega_1, \omega_2) d\omega_1 d\omega_2$$

Local averaging of stationary random functions

For a stationary random function $Z(t)$, the local moving averaging function $Z_T(t)$ of $Z(t)$ is defined as

$$Z_T(t) = \frac{1}{T}\int_{t-T/2}^{t+T/2} Z(\tau)d\tau$$

The figure below shows the Local (moving) averaging of a stationary random function.



Local averaging of stationary random functions

Note that local averaging of a stationary random process will not affect the mean, but it does reduce the variance. So the variance of the averaged process $Var[Z_T(t)] = E\left[(Z_T(t) - \mu_Z)^2\right]$ can be calculated for $\mu_Z = 0$ as

$$\begin{aligned}
Var[Z_T(t)] &= E\left[\frac{1}{T}\int_{t-T/2}^{t+T/2} Z(\tau_1)d\tau_1 \frac{1}{T}\int_{t-T/2}^{t+T/2} Z(\tau_2)d\tau_2\right] \\
&= E\left[\frac{1}{T}\int_{0}^{T} Z(\tau_1)d\tau_1 \frac{1}{T}\int_{0}^{T} Z(\tau_2)d\tau_2\right] \\
&= \frac{1}{T^2}\int_0^T\int_0^T E\left[Z(\tau_1)Z(\tau_2)\right] d\tau_1 d\tau_2 \\
&= \frac{1}{T^2}\int_0^T\int_0^T C_Z(\tau_2 - \tau_1) d\tau_1 d\tau_2
\end{aligned}$$

From $Var[Z_T(t)]$ a new function is introduced called the **variance function**

$$V_Z(T) = \frac{Var[Z_T(t)]}{\sigma_Z^2}$$

The variance function thus describes the reduction in variance when averaging a random function as a function of the averaging interval T.

Local averaging of stationary random functions

The variance function is related to the correlation function as

$$V_Z(t) = \frac{1}{T^2} \int_0^T \int_0^T \rho_Z(\tau_2 - \tau_1) d\tau_1 d\tau_2$$

This equation can be simplified as

$$V_Z(t) = \frac{1}{T} \int_0^T \left(1 - \frac{\tau}{T}\right) \rho_Z(\tau) d\tau$$

The table below gives a number of correlation functions and their variance functions.

Exponential (first order autoregressive process)	$\rho_z(\tau) = e^{-\tau/a}$ for $\tau \geq 0,\ a > 0$	$V(T) = 2\left(\frac{a}{2}\right)^2 \left(\frac{T}{a} - 1 + e^{-T/a}\right)$ $\theta = 2a$
Second order autoregressive process	$\rho_z(\tau) = \left[1 + \frac{\tau}{a}\right] e^{-\tau/a}$ for $\tau \geq 0,\ a > 0$	$V(T) = \frac{2a}{T}\left[2 + e^{-T/a} - \frac{2a}{T}\left(1 - e^{-T/a}\right)\right]$ $\theta = 4a$
Gaussian correlation function	$\rho_z(\tau) = e^{-\tau^2/a^2}$ for $\tau \geq 0,\ a > 0$	$V(T) = \left(\frac{a}{T}\right)^2 \left[\sqrt{\pi} Erf\left(\frac{T}{a}\right) - 1 + e^{-T^2/a^2}\right]$ $\theta = a\sqrt{\pi}$

where $\tau = |t_2 - t_1|$.

Local averaging of stationary random functions

If we analysed the variance function when $T \to \infty$, $\lim_{T\to\infty} V_Z(T) \to \frac{\theta}{T}$, where θ is the **scale of fluctuation**. The scale of fluctuation is also given for the three correlation models given in the table above. The scale of fluctuation was already introduced earlier as a measure of spatial correlation of the random function and can also be calculated using the correlation function or the spectral density.
$V_Z(T) \to \frac{\theta}{T}$ states that for larger averaging intervals the variance reduction through averaging is inversely proportional to the scale of fluctuation: the larger the scale of fluctuation the larger T should be to achieve a given variance reduction.
The covariance of the averaged process is given by

$$C_Z(T, \tau^2) = \frac{1}{T^2} \int_0^T \int_0^{\tau+T} C_Z(t_1, t_2) dt_1 dt_2$$

This equation is usually solved by the implementation of numerical integration.
In the case of 2D random fields (in space where $\mathbf{x} = x, y$), the local average process is over an area $A = L_1 L_2$ and is defined as

$$Z_T(x_1, x_2) = \frac{1}{L_1 L_2} \int_{x_1 - L_1/2}^{x_1 + L_1/2} \int_{x_2 - L_2/2}^{x_2 + L_2/2} Z(u_1, u_2) du_1 du_2$$

Local averaging of stationary random functions

The variance function is given by

$$V_Z(L_1, L_2) = \frac{1}{L_1 L_2} \int_0^{L_1} \int_0^{L_2} \left(1 - \frac{h_1}{L_1}\right) \left(1 - \frac{h_2}{L_2}\right) \rho_Z(h_1, h_2) dh_1 dh_2$$

The limit of the variance function defines the spatial **scale of fluctuation** or **characteristic area**

$$\lim_{L_1, L_2 \to \infty} V_Z(L_1, L_2) \to \frac{\alpha}{L_1 L_2}$$

where

$$\alpha = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \rho_Z(u_1, u_2) du_1 du_2$$

The characteristic area α can also be obtained through the spectral representation by setting $\omega_1 = \omega_2 = 0$ in the Wiener-Khinchine relation, so

$$s_Z(0,0) = \frac{S_Z(0,0)}{\sigma_Z^2} = \frac{1}{4\pi^2} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \rho_Z(h_1, h_2) dh_1 dh_2$$

combining the two last equations

$$\alpha = 4\pi^2 s_z(0,0)$$

Local averaging of stationary random functions

The covariance of the averaged random field in 2D is given by

$$C_Z(L_1, L_2; h_1, h_2) = \frac{1}{(L_1 L_2)^2} \int_0^{L_1} \int_0^{L_2} \int_{h_1}^{L_1+h_1} \int_{h_2}^{L_2+h_2} C_Z(x_1, y_1, x_2, y_2) dx_1 dy_1 dx_2 dy_2$$

The covariance of the spatially averaged random function is frequently used in geostatistical mapping, where its values are approximated with numerical integration. In vector notation where $\mathbf{x_1} = (x_1, y_1)^T$, $\mathbf{x_2} = (x_2, y_2)^T$, $\mathbf{h} = (h_1, h_2)^T$ and $A = L_1 L_2$

$$C_Z(A, \mathbf{h}) = \frac{1}{A^2} \int_A \int_{A+\mathbf{h}} C_Z(\mathbf{x_1}, \mathbf{x_2}) d\mathbf{x_1} d\mathbf{x_2}$$